CORTEX CODE

SkyTrace AI

Visualizing uncertainty.
Guiding the search.

A hackathon prototype for exploring illustrative search areas from last-known position and flight details.

MORE CLARITY · BETTER DECISIONS · GREATER HOPE



GEO-SPATIAL HACKATHON

PROTOTYPE DEMONSTRATION | NOT FOR OPERATIONAL SAR

01 — PROBLEM

When an aircraft goes missing, uncertainty expands.

A search can begin with incomplete information.

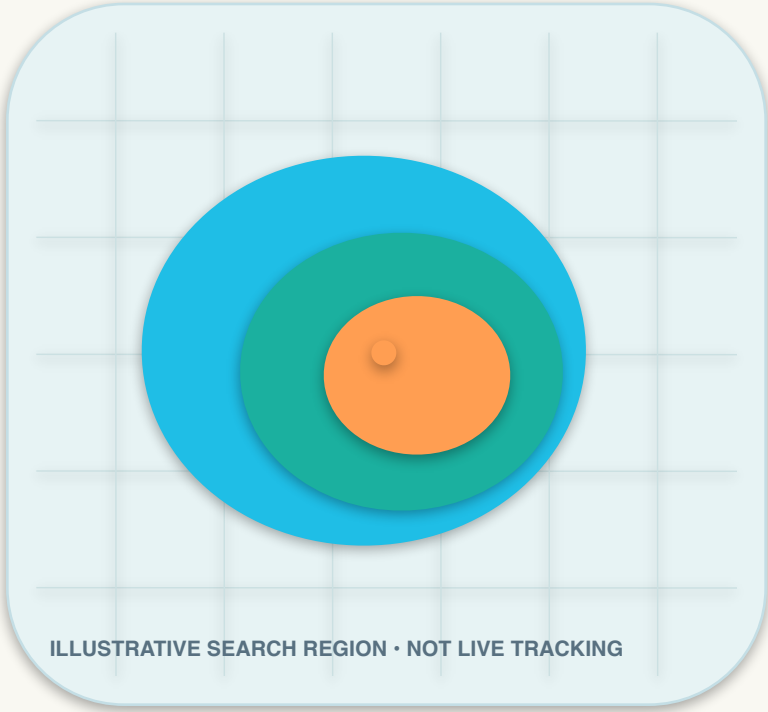
Teams may need to reason from a last-known position, elapsed time, heading and speed—while the true location remains unknown.

1 POSITION UNCERTAINTY

Last-known coordinates may not reflect the aircraft's current position.

2 TIME PRESSURE

A broad area is difficult to interpret quickly.



Conceptual view — not real flight data

02 — PROPOSED SOLUTION

Turn flight details into a visual search-area sketch.

SkyTrace AI helps users explore scenarios.

Enter a last-known position and selected flight details. The prototype generates an illustrative region and demo grid that make assumptions easier to see.

A INPUT

Position + flight details

B SCENARIOS

Heading sensitivity

C VISUALIZE

Region + demo grid



03 — PROJECT OVERVIEW

Built to make assumptions visible—not hide them.

Interactive map

See the illustrative region in a geographic context.

Scenario comparison

Explore how heading assumptions can change the sketch.

Demo grid

Inspect a relative, demonstration-only grid.

Explainable limits

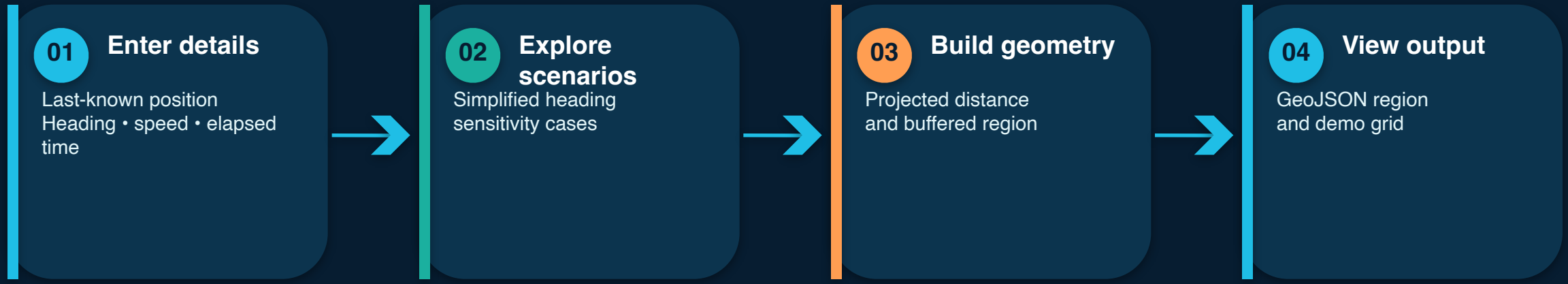
Show the assumptions and caveats beside the output.

DESIGN PRINCIPLE

A clear visual aid for reasoning about uncertainty—not a prediction of an aircraft's actual location.

04 — WORKING PRINCIPLE

A simple path from inputs to an illustrative map.



Every output depends on simplified assumptions and should be interpreted as illustrative.

05 — INPUTS & ASSUMPTIONS

Make the inputs and their meaning explicit.

01 Last-known position

Starting geographic reference

02 Heading / ground track

Scenario direction, not guaranteed path

03 Ground speed + elapsed time

Baseline distance estimate

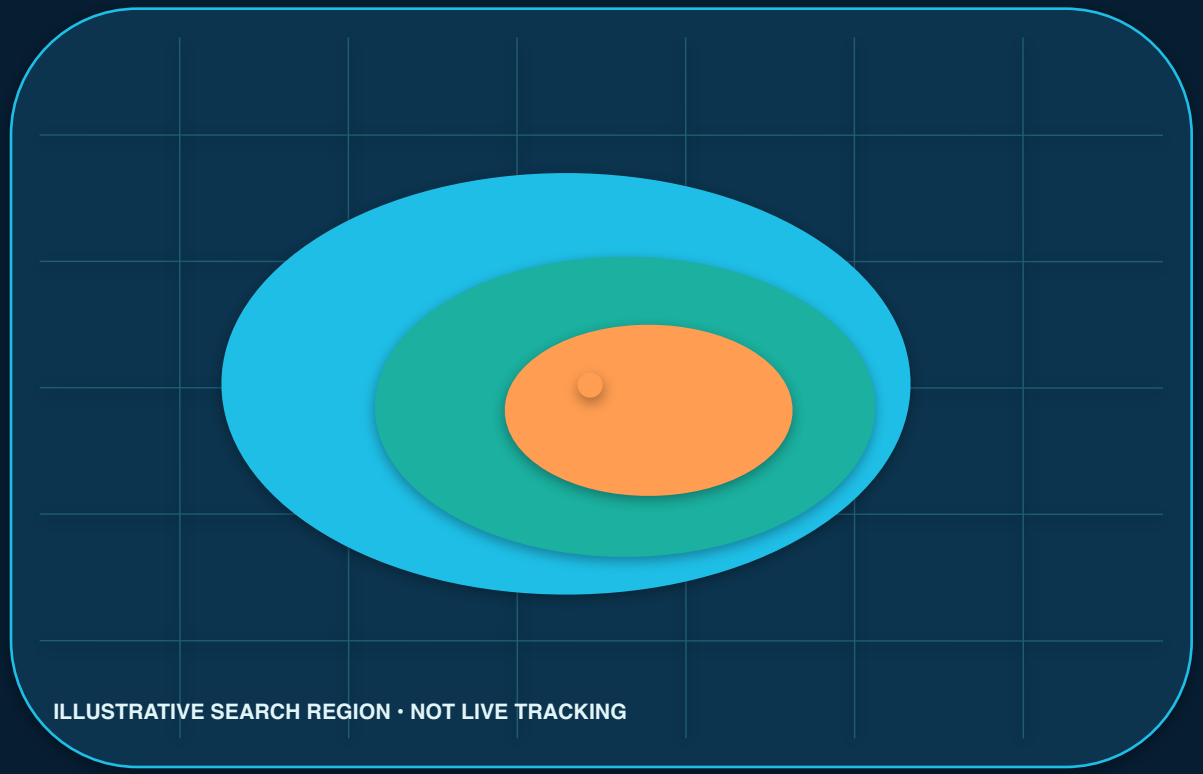
04 Wind + fuel context

Contextual inputs; not full drift or endurance modeling

Wind is a hypothetical scenario; fuel percentage is contextual and does not estimate endurance.

06 — MAP & RESULTS

A dashboard view for exploring a hypothetical region.



01 Search region

Buffered illustrative geometry

02 Heading cases

Compare sensitivity scenarios

03 Demonstration grid

Relative scores are for visual exploration only; they are not probabilities.

No live aircraft tracking or operational statistics.

07 — TECHNICAL ARCHITECTURE

A lightweight geospatial stack.

UI Frontend

React • TypeScript • Vite
Tailwind CSS • Leaflet

API API backend

FastAPI • Pydantic
REST endpoints

GEO Geospatial engine

PyProj • Shapely
WGS84 / AEQD projection

REQUEST FLOW

React UI → FastAPI → geometry calculations → GeoJSON + explanatory response

Runs locally; the prototype does not require paid AI APIs or a database.

08 — GEOSPATIAL METHOD

Simple geometry, visible uncertainty.

BASELINE DISTANCE

distance = speed × time

Ground speed × elapsed hours

Heading sensitivity

Alternate directions form scenario envelopes.

Buffered hull

A simplified region visualizes possible spread.

This is not a probabilistic model, validated drift simulation, or real-world location estimate.

09 — FEASIBILITY & IMPACT

A realistic prototype with a clear next-step path.

FEASIBILITY

- Uses established open-source libraries
- Local API and map-based interface
- No database or paid API required
- Inputs and assumptions are inspectable

POTENTIAL VALUE

- Makes scenario assumptions easier to discuss
- Provides a visual aid for demonstrations
- Supports prototyping of geospatial workflows
- Offers a foundation for expert-led validation

Any real-world use would require domain experts, validated data, testing, and operational approval.

10 — SAFETY & LIMITATIONS

A prototype is a tool—not a rescue decision.

!

Important boundaries

- Does not locate or track real aircraft.
- Uses simplified, illustrative assumptions.
- Demo grid scores are not probabilities.
- Does not model terrain, ocean currents, debris drift, or survival.
- Not a replacement for official search-and-rescue operations.

FOR DEMONSTRATION AND EDUCATIONAL EXPLORATION ONLY

11 — TEAM

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THINK • BUILD • SOLVE

More clarity.

Better decisions.

Greater hope.

Visualizing uncertainty.
Guiding the search.



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THANK YOU